MTA Network Fundamentals Study Sheet

Unicast - Assigned to a ***single*** **network interface** located on a specific subnet on the network and used for *one to one communication*.

Multicast - Assigned to the ***variable*** **portions of an IPv4 address** that is used to identify a network node's interface on a subnet. *(One to Many)*

- Used by IPv6
- Used only by IPv4 Class D

Broadcast - **Assigned to all network interfaces** located on a subnet on the network and used for *one to every* on a subnet communication.

Anycast - One to nearest communications.

IP Addressing:
Private Network: First octet starts with 10, 172 or 192
Multicast Address : 224.0.0.0 - 239.255.255.255 (Class D)
Loopback Address: 127.0.0.0 - 127.255.255.255

NAT - Network Address Translation:

- Resolves (Converts / Translates) Private IP to Public IP & vice versa

IP Classes Ranges:
Class A: 1-126
Class B: 128-191
Class C: 192 - 223
Class D: 224 - 239 (IPv4 Multicast)
Class E: 240 - 255 (Reserved for Govt.)

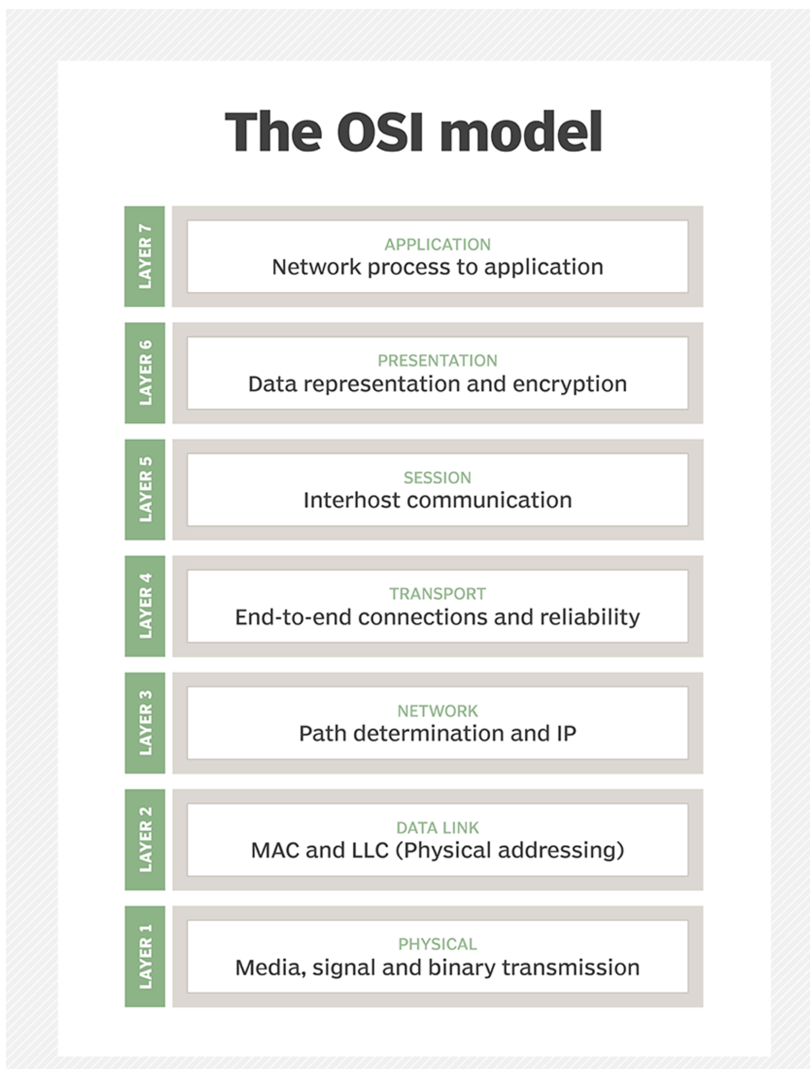
Network ID ends with = 0

- Ex: 192.183.19.0

Broadcasting ID ends with = 255

- Ex:192.183.19.255

Loopback Address: test NIC card.



There are 32 bits in an IPv4 address.
There are 128 bits in an IPv6 address.

169 Addresses = APIPA
Default Subnet Masks
Class A: 255.0.0.0
Class B: 255.255.0.0
Class C: 255.255.255.0

- Subnets = subnet network
- Subnet Mask determines which portion is the network addy and what is the host.

OSI: KEY WORDS
Application Layer (7)
Emails (SMTP, POP3, IMAP), HTTP, FTP, DNS, **Directory services & remote file services (RDP)**
(as soon on test^^)

Presentation Layer (6)
Operating System (MAC, Windows, Linux) & Encryption (!!)

Session Layer (5)
Connection between two endpoints

Transport Layer (4)
Segments, Checks for errors in data once it arrives
TCP: Uses Three way handshake to establish a connection.
UDP: Unreliable Question

Networking Layer (3)
Packets, Router, Layer 3 Switch

Data Link Layer (2)
MAC, LLC, Frames, Switches, hubs, bridges

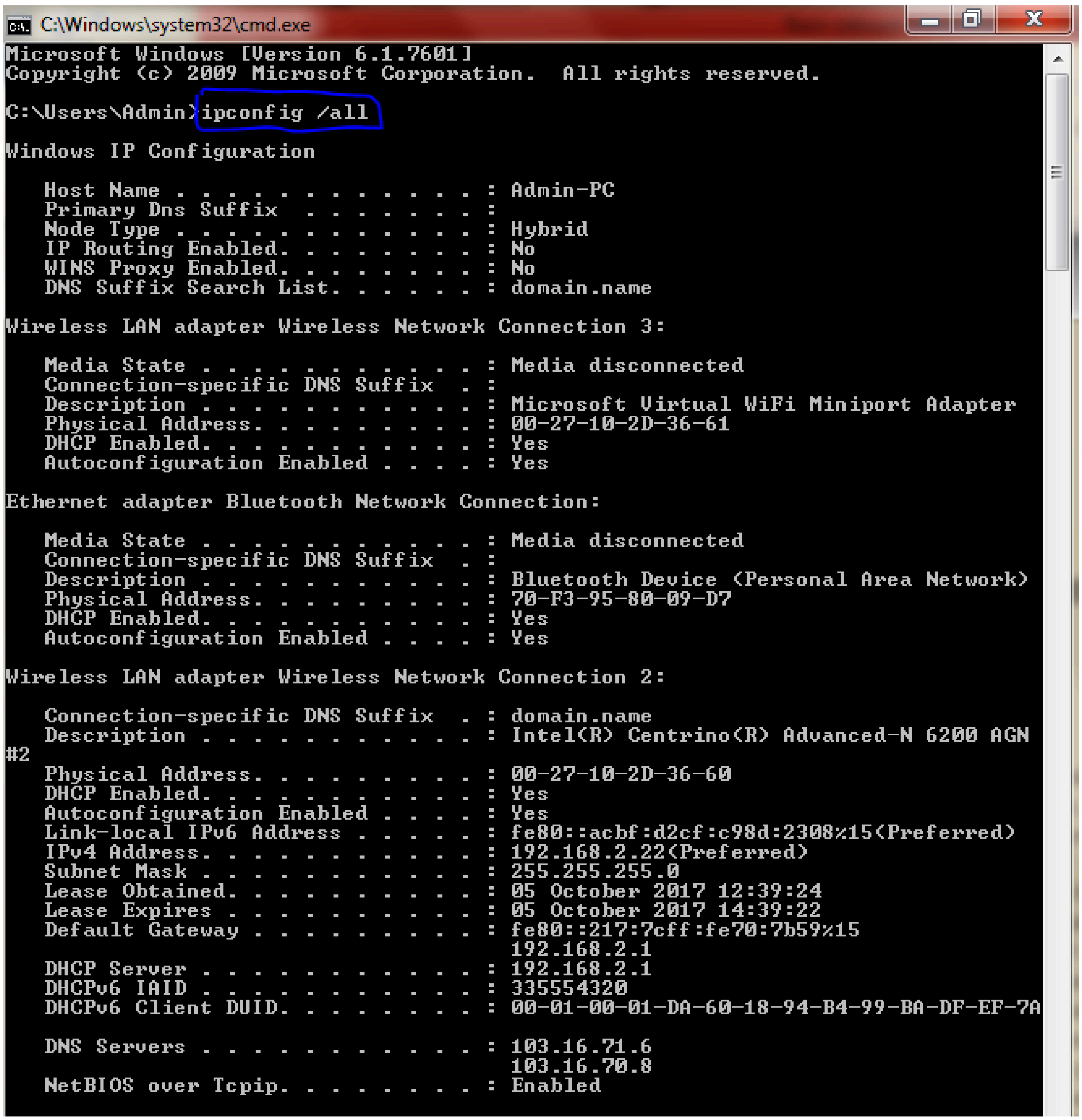
Physical Layer (1)
Bits, binary, cables, Media (connections)

Commands:

Ipconfig /release + ipconfig /renew - forces a client (host) to renew its address lease from a DHCP server.

DHCP - Automatic assignment of IP addresses

When you are reading a command screen, look at the connection they ask for. (ex: Wireless LAN connection, Ethernet Adapter, etc.) ipconfig command will show result for all network connections on a device if they are asking if DHCP is enabled for the Wireless Lan, answer is Yes.



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C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
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C:\Users\Admin>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Admin-PC
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : domain.name

Wireless LAN adapter Wireless Network Connection 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Virtual WiFi Miniport Adapter
   Physical Address. . . . . . . . . : 00-27-10-2D-36-61
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 70-F3-95-80-09-D7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wireless Network Connection 2:

   Connection-specific DNS Suffix  . : domain.name
   Description . . . . . . . . . . . : Intel(R) Centrino(R) Advanced-N 6200 AGN
#2
   Physical Address. . . . . . . . . : 00-27-10-2D-36-60
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::acbf:d2cf:c98d:2308%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.2.22(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 05 October 2017 12:39:24
   Lease Expires . . . . . . . . . . : 05 October 2017 14:39:22
   Default Gateway . . . . . . . . . : fe80::217:7cff:fe70:7b59%15
                                       192.168.2.1
   DHCP Server . . . . . . . . . . . : 192.168.2.1
   DHCPv6 IAID . . . . . . . . . . . : 335554320
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-DA-60-18-94-B4-99-BA-DF-EF-7A

   DNS Servers . . . . . . . . . . . : 103.16.71.6
                                       103.16.70.8
   NetBIOS over Tcpip. . . . . . . . : Enabled
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Ad Hoc Network *(considered a topology like star, bus and mesh)*

- An ad hoc network is a type of **temporary computer-to-computer connection**. (peer to peer P2P / PtP / PP) In ad hoc mode, you can set up a wireless connection directly to another computer without having to connect to a Wi-Fi access point or router.
- More than one laptop can be connected to the ad hoc network, as long as all of the adapter cards are configured for ad-hoc mode and connect to the same SSID. The computers need to be within 100 meters of each other.
- Does not use WEP, **WPA2**, WPA
- Adhoc is more secure than WAP

Commands:

Tracert = Trace Route, traces route of packets

Pathping = combines the functionality of ping with that of tracert. It is used to locate spots that have network latency and network loss.

Ping = used to test continuity (connection) between two nodes on a network.

Wireless Protocols

WEP = Weakest form of wireless encryption

WPA = Uses TKIP encryption

WPA2 = Strongest form of encryption (AES is the form of encryption). (WPA2 Enterprise is the strongest form)

Switch vs Hub:

- **Switch:** are capable of sending and receiving data at the same time. & identify the intended destination of the data they receive.
- **Hubs**: cause more **data collisions than switches** & send each packet to all the computers that are connected to them.

Switch vs Bridge

- Bridges or switches are used to connect these subdivided segments of networks. In a long way, the terms bridge and switch are used interchangeably.
- <u>Bridge and switch both provide the same functionality</u> but the switch does it with greater efficiency.

Layer 3 / Multilayer Switch = Provides Layer 3 routing functions. (A regular switch is dealt with in layer 2 unlike a Multi Layer Switch which can also be used in a Layer 3 (Networking))

DNS

DNS: **"**Resolves" (Finds / Matches) Domain names to IP Addresses.
So when we type in "google.com" on our web browser, the computer sees it as "64.233.160.0"
The DNS's job is to provide users with a "user friendly name"

Recursive:

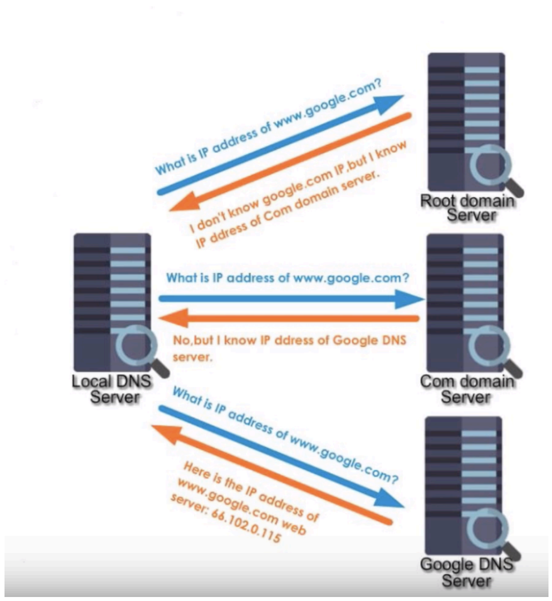
- Google Definition:
  Recursive DNS queries occur when a DNS client requests information from a DNS server that is set to **query subsequent DNS servers until a definitive answer is returned to the client**. The queries made to subsequent DNS servers from the first DNS server are iterative queries
  Best way to remember Recursive query is to memorize that *burden is on Server to resolve the query*.
- Ms. Akhter's Version:
  Your Local PC asks local DNS Server for a direct answer. The request is considered recursive because the server provides your PC with a direct answer. The answer can be the IP address of the domain you are trying to visit or it can be "I Dont Know"

Ittiritive:

- Google's Definition:
  Iterative DNS queries are ones in which a DNS server is queried and returns an answer without querying other DNS servers, even if it **cannot provide a definitive answer**. Iterative queries are also called non-recursive queries.
  Best way to remember an Iterative query is to memorize that *burden is on Client to resolve the query*.

- Ms. Akhter's Version:

  Iterative DNS asks higher level DNS servers (ROOT DNS, TLD, etc.) for answers. In the Iterative process, requests are constantly "iterated" (asked over and over) by local DNS servers to higher DNS servers until an answer is found.



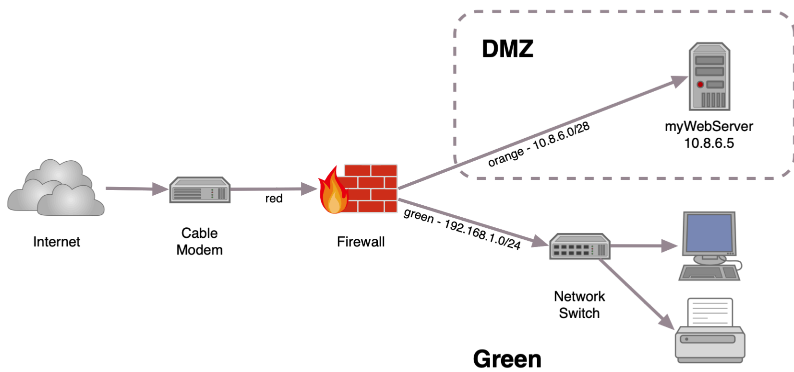
QoS = Quality of Service

Mesh Topology = Every node connects to every other node on the network. & It is fault tolerant because of redundant connections.
The INTERNET is a MESH topology.

Other Facts:

- **Network Firewall** protects a network's perimeter by monitoring traffic as it enters and leaves. (using ports and protocols)

- A network that separates an organization's private network from a public network is a **PERIMETER**.



A **DMZ (Demilitarized Zone Network)** also sometimes called Perimeter Network is used for network security. A DMZ logically separates a newtork and is used for web servers, e-mail servers, FTP servers and other public servers to gain access from or to the internet.
Placing public facing servers in a DMZ separates it from its the LAN which you DONT want to be as easily accessible by the public. DMZs logically protects your internal network.

- **100BaseTX** networks are wired together in a star topology using unshielded twisted-pair (UTP) cabling or shielded twisted-pair (STP) cabling and 100-Mbps hubs or Ethernet switches. If UTP cabling is used (which is the most common scenario), it must be category 5 cabling (cat5 cabling)
- **1000Base- T** (Four pairs of Category 5 UTP cable for a maximum length of 100 meters.

- IPv4 routers (CAN NOT) forward network broadcast outside of the local network subnet. (this is true, so you select Yes)
- IPv6 Traffic can be transported between IPv6 networks over an IPv4 network (Select Yes)
- When you configure Windows computers to obtain an IP address automatically but cannot contact a DHCP server, they will assign an APIPA Address. (Select Yes)

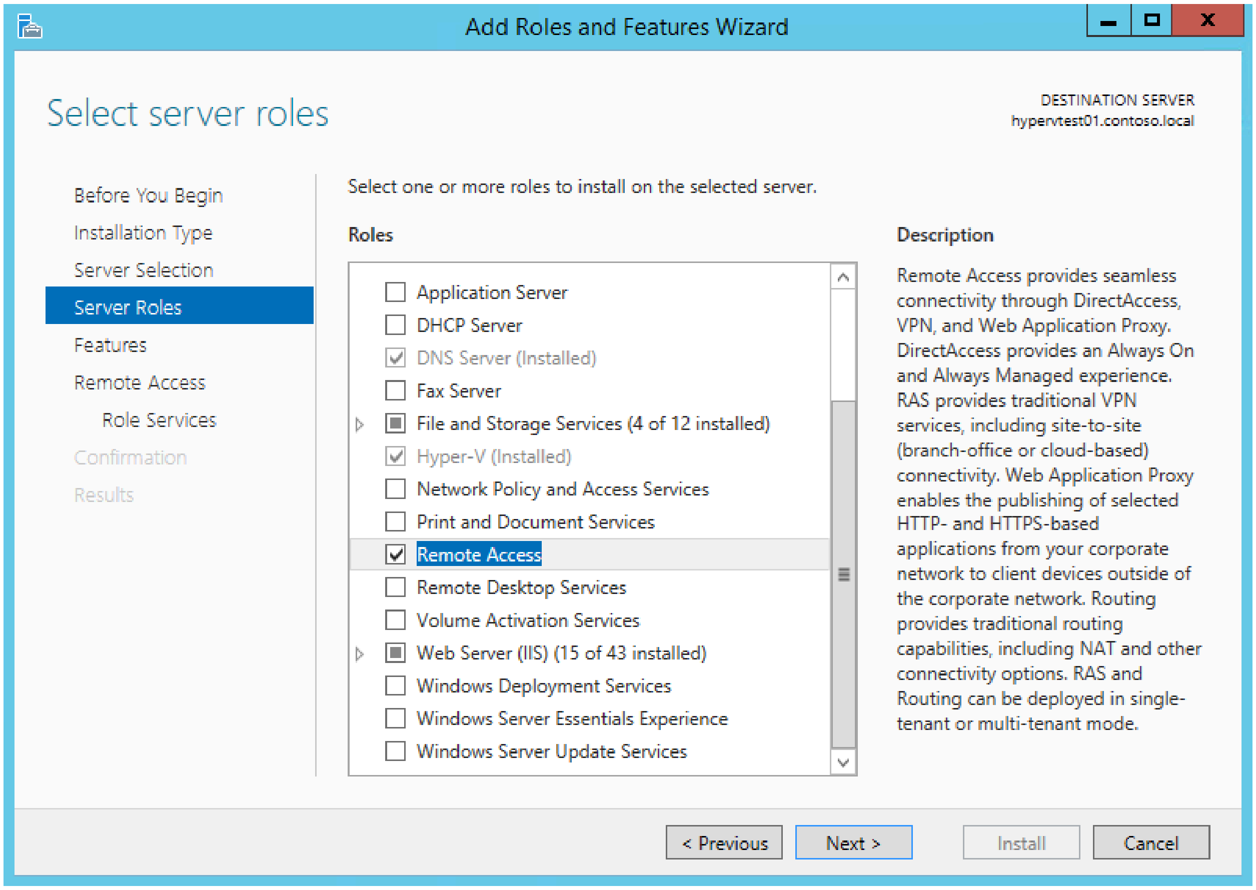
- **APIPA ADDRESSES ALWAYS START WITH 169 (First Octect)**
  - APIPA = Automatic Private IP Addressing
  - Your PC receives an APIPA Address only when the device is unable to connect the DHCP client or when DHCP fails / is unavailable.

- **Windows Internet Name Service (WINS)** is a service that resolves NetBIOS names to IP addresses in routed networks.

- A **Hosts file** is a file that almost all computers and operating systems can use to map a connection between an **IP address and domain names (DNS)**.

- Ping uses ICMP

- Interactive Lab: Windows Server Screen - Answer: check off “Remote Access” The lab will ask you to give client remote access. You will have to use the sidebar to scroll down and check off “remote access.”



Ports:
DNS = 53
FTP = 20 & 21
LDAP = 389
SNMP = 161
RDP = 3389
Telnet = 23
SSL / TLS = 443
HTTPS = 443
HTTP = 80
IMAP = 143
SSH = 22
POP3 = 110

SMTP = 25

RIP - Routing Information Protocol
15 max hop count - 16 is considered as network unreachable

ARP = IP Address to HOST NAME (MAC Address)
STP = Preventers interference

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